Walnut Technologies™

COMPANY PRESENTATION 2026

Intelligent Embedded Systems, Engineered for a Connected World

End-to-end design, development and manufacturing partner (ODM/OEM) for medical devices, neuro-rehabilitation robotics, Payment system and its software eco-system, IoT and automotive electronics.

Walnut Technologies™ · Mohali, India · walnutmedical.in



Who we are

Walnut Medical & Walnut Technologies, established in 2016, has grown into a vertically integrated design and manufacturing company — from PCB design to final assembly, every stage is in-house.

10+	**600+**	**10+**
Years of excellence	Engineers & technicians	Countries served
500K+	**150,000**	**4**
Units manufactured / month	Sq. ft facility	SMT production lines

Mohali · AMTZ Visakhapatnam · Dhanana (Upcoming in Haryana)

ISO 13485 · CE · FCC · BIS · Industry 4.0 · Full traceability


Production line


SMT line · JUKI & FUJI

Vision, mission & strengths

OUR VISION

To become India's leading technology and manufacturing powerhouse, delivering innovative, high-quality payment and electronic solutions to global markets.

OUR MISSION

Customer First

Exceed expectations with innovative, reliable solutions.

Quality Excellence

Superior products through continuous improvement.

Integrity & Trust

The highest standards of ethics and transparency.

Sustainable Growth

Lasting value for customers, partners and stakeholders.

R&D

Research & Development

Dedicated research and innovation driving next-gen hardware, firmware, and embedded system technologies.

D&D

Design & Development

In-house design capability — from industrial aesthetics and mechanical CAD to schematics, PCB layout, and validation.

Cloud & Firmware

In-House Architecture

AWS server-side, RTOS platforms, MQTT, HTTPS, FOTA and JSON for connected IoT devices.

Lean Mfg.

Six Sigma & 5S

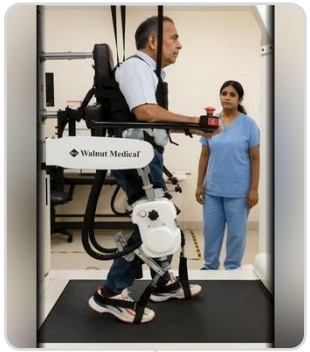
5S, Kaizen and Six Sigma practices — less waste, shorter lead times, and consistent high quality.

Fast Growth

100%+ CAGR

CAGR above 100% over the last four years with expanding international footprint.

Our journey











2016
Walnut Medical founded

Started with a vision to make quality healthcare more accessible.

2017
Robotics & neuro-rehab

Walkex functional therapy and the WalkLab gait-training robot.

2019
Medical devices

IR thermometer and digital BP monitor, then nebulizers and oxygen concentrators.

2021
Walnut Technologies

Engineering expands beyond healthcare into electronics for every industry.

2022
Fintech · soundbox

Smart payment soundboxes and POS devices, in mass production from 2023.

2025
Smart clusters

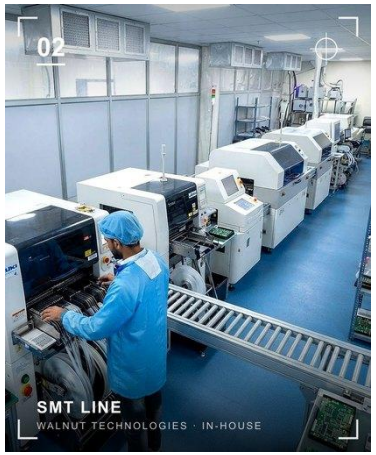
5″ and 7″ smart clusters for 2-wheelers, tractors and vehicles.

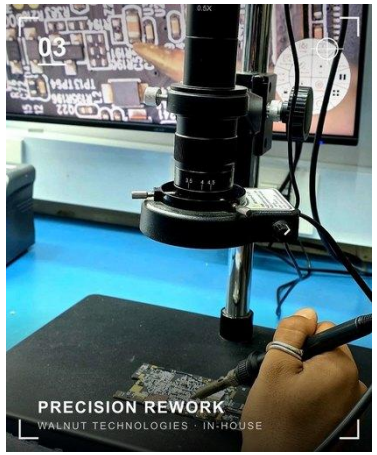
2026
GPS tracker & IoT

GPS tracker & gateway and a growing range of connected IoT devices.

Manufacturing infrastructure













150,000 sq. ft facility
Mohali · AMTZ · Dhanana (Upcoming in Haryana)

4 SMT production lines
Fully automatic SMT — JUKI RS-1, FUJI, 200,000 CPH

In-house test & calibration
Keysight · National Instruments · Rigol · Tektronix

PCB design to production
Design, validation, prototyping, pilot lots, full scale

IQC · LQC · OQC quality teams
Inspection at incoming, line and outgoing stages

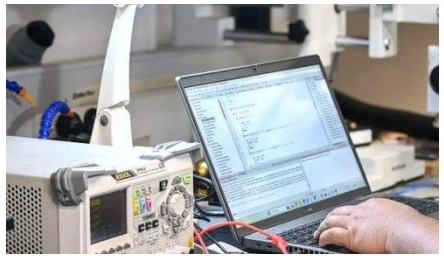
Moulds & injection moulding
Mass-production mould design and plastics in-house

End-to-End Production Process



01 Idea & Requirement

Clear direction



02 Research & Market Analysis

Right opportunity



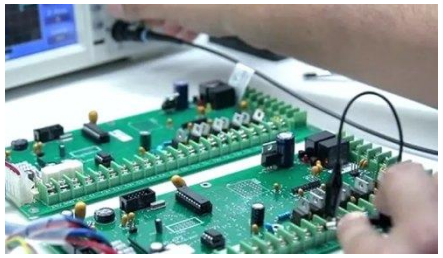
03 Design & Engineering

Optimized design



04 Prototype Development

Working prototype



05 Validation & Testing

Validated performance



06 Manufacturing Engineering

Production ready



07 Quality Assurance

Zero-defect mindset



08 Packaging & Dispatch

Safe, on-time delivery



09 After-Sales Support

Reliable support



10 Continuous Improvement

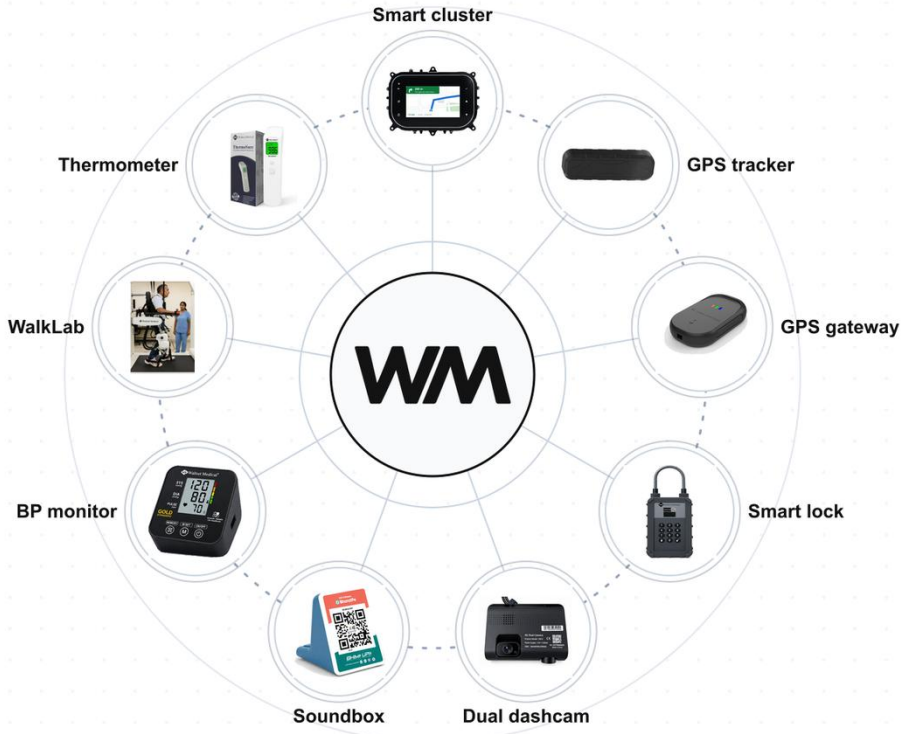
Better every day

SECTION 04

Products

Five product families, designed and built in-house.

Medical	Diagnostics & monitoring
Neuro-rehabilitation	Robotics & FES therapy
Fintech	Soundboxes & payment POS
Automotive	Smart instrument clusters
IoT	Trackers, gateways & locks



Medical devices

CDSCO-approved Class B and Class C medical devices, designed and manufactured under ISO 13485.



Digital BP monitor

Clinical-grade automatic blood pressure monitor with digital display and memory.



IR thermometer

ThermoSure non-contact infrared thermometer — fast, hygienic readings.



Compressor nebulizer

Compact, low-noise respiratory therapy for clinic and home.



Oxygen concentrator

5 L and 10 L, single and dual flow, for clinical and homecare use.

In development: a gold-standard electronic BP monitoring device, with ICMR funding and clinical trials at AIIMS Nagpur, AIIMS Delhi and Chettinad Super Speciality Hospital.

Neuro-rehabilitation & robotics

AI-assisted therapy | Real-time monitoring | Tele-rehab ready | FES | rTMS | RMNS



WalkLab

Lower limb walk trainer.



Weebo

Early rehabilitation robot and functional electrical stimulation.



Walkex

Functional electrical stimulation for foot drop — stroke and paralysis recovery.

Fintech & payment devices

BIS approved | PCI-DSS | NPCI compliant | EMV L1/L2 | Multi-language audio | NFC tap & pay — 2026





All in one

On-device QR generation with EMV tap and chip insert, and a certified PIN pad.



DQR — Digital QR

High-contrast LCD for dynamic QR and clear transaction visibility.



LDQR — Large display

10″ touchscreen, 2.4″ merchant display, keyboard and NFC tap-and-pay.



Payment soundbox

4G/2G, NB-IoT, Bluetooth and Wi-Fi smart payment speakers.

Terminal Management System

Manage your fleet anywhere, anytime. A control centre for your entire terminal estate — TMS streamlines the full device lifecycle, from deployment to OTA updates and troubleshooting, with real-time visibility into every unit.

Device & network

Storage, CPU, memory, battery, IP, IMEI, operator and live location.

App details

App, SDK, sound-file and language versions; install history.

Scheduler

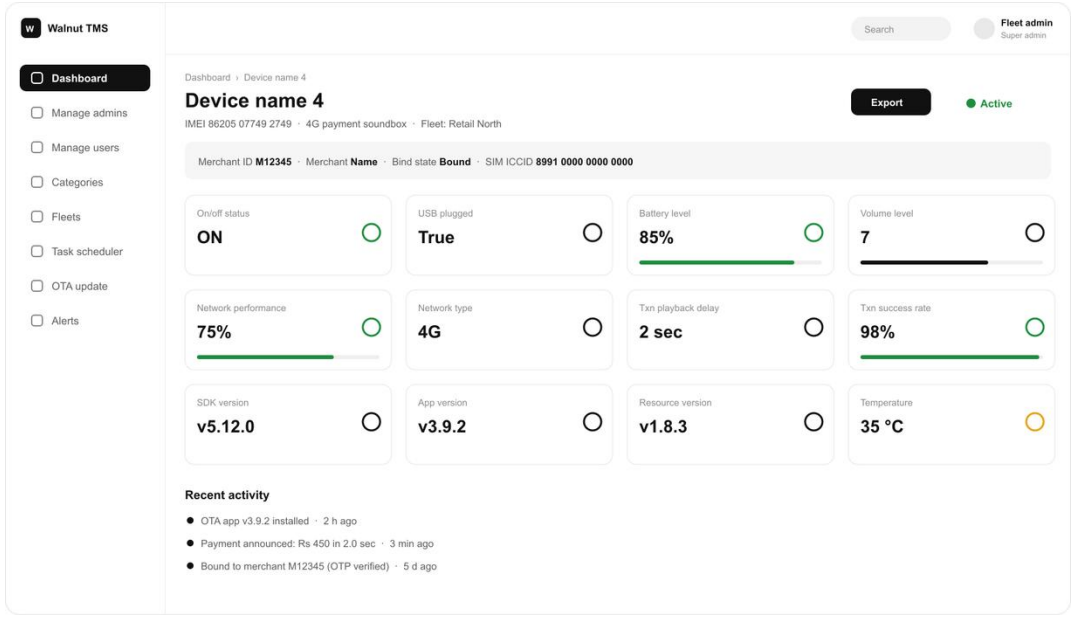
Schedule OTA jobs, advertisements and remote commands.

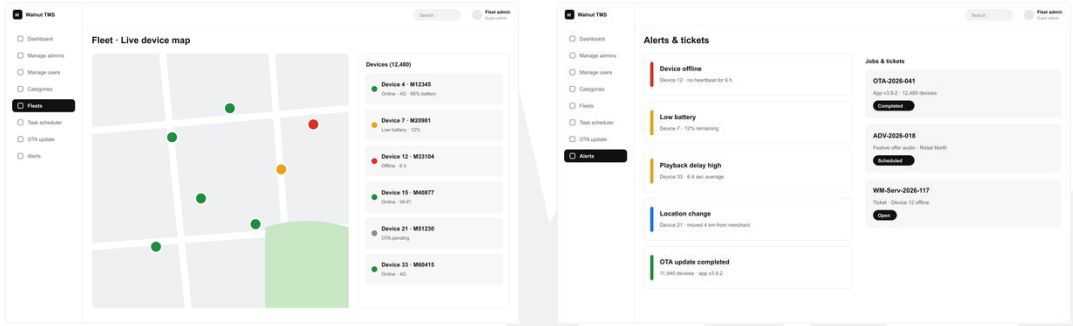
Onboarding

Link to merchant accounts, release OTP, un-bind, watch the transaction matrix.

Defender

Security metrics, anomaly detection, audits, alerts and mitigation.





MQTT broker — instant, secure announcements

Carrier-grade, isolated messaging infrastructure delivering payment voice confirmations in under 200 ms with end-to-end cryptographic integrity.

Dedicated connection

MQTT traffic isolated on its own fast-path to prevent network congestion and guarantee ultra-low latency.

TLS 1.2 encryption

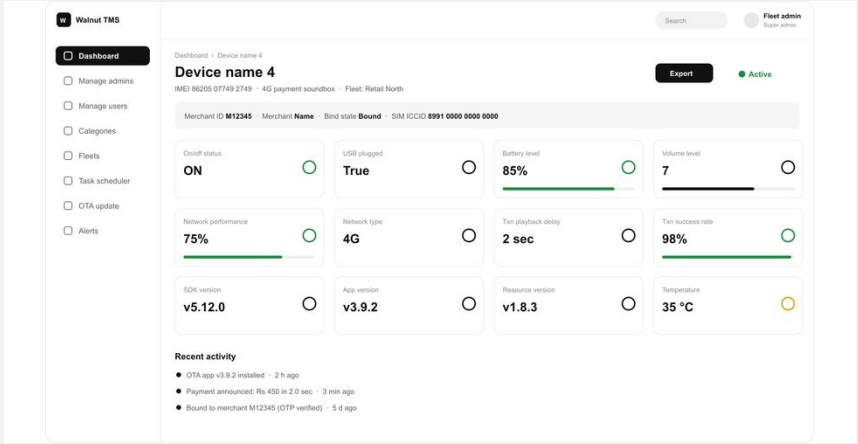
Hardware-accelerated end-to-end encryption, MITM protection, and verified data integrity in transit.

Centralised broker

Streamlined broker architecture for simplified access control, ACLs, and enterprise policy enforcement at scale.

Lightweight by design

Minimal packet overhead, QoS 1 delivery, and optimized keep-alive intervals tailored for battery-operated soundboxes.



< 200 ms
Voice announcement latency

TLS 1.2
Bank-grade payload encryption

99.99%
Delivery success rate

Automotive & IoT



Smart clusters

5″ and 7″ · Android & RTOS · 4G, BT, Wi-Fi · Google Maps mirroring · custom cluster development



GPS tracker

WM-MFT-4G · <2.5 m accuracy · 4G CAT-1 · geo-fencing · ignition monitoring · SOS



GPS gateway

Asset tracking and telemetry · fleet management · secure cloud connectivity



4G smart lock

Remote access control · real-time status · IP67 · with battery backup

Smart instrument clusters







Next-generation digital clusters with high-resolution displays, real-time vehicle data and customisable driver interfaces — built for durability and clarity.

5-inch and 7-inch displays
Two sizes to suit compact and full-size dashboards

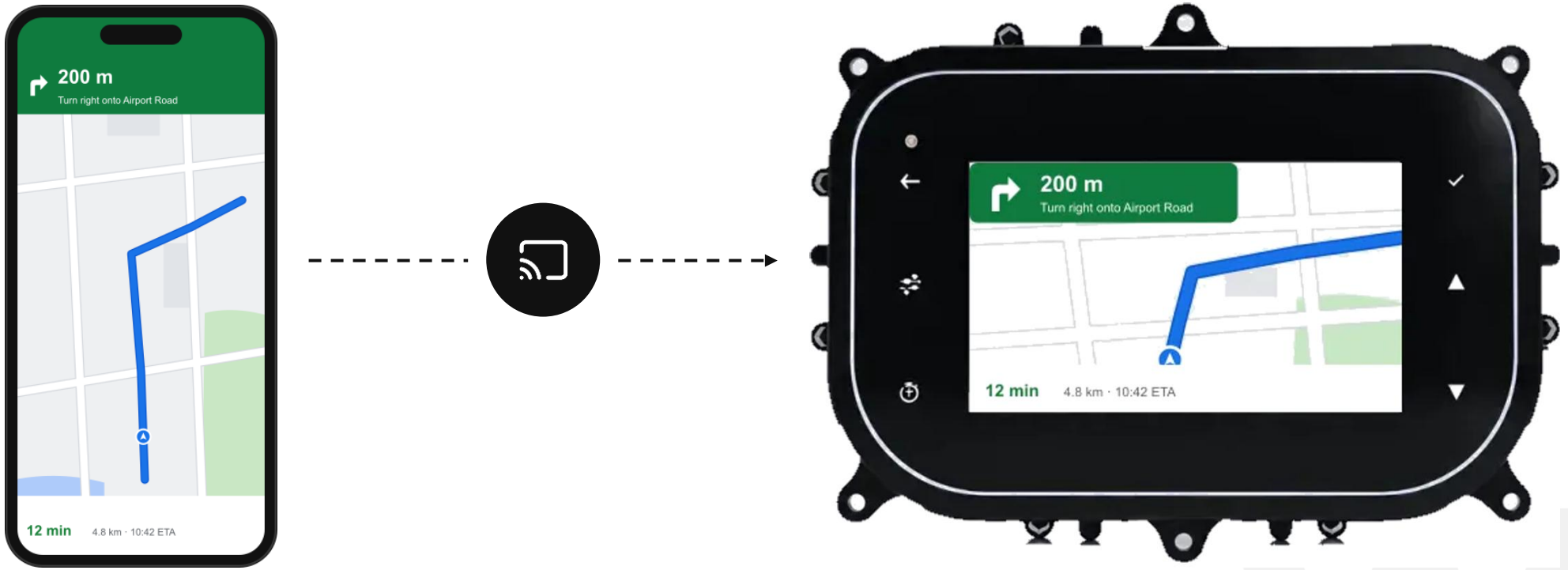
Android and RTOS based
Smart and cost-optimised platform options

4G · Bluetooth · Wi-Fi
Connected cluster with phone screen mirroring

BUILT FOR

- 2-wheelers
- Tractors & farm equipment
- LMV & HMV

Google maps capture on non-andriod cluster



Turn-by-turn on the dash

Google Maps navigation from the rider's phone appears on the cluster screen.

Eyes stay on the road

Directions sit in the driver’s line of sight — no phone on the handlebar or dashboard.

Works with your phone

Pair over the cluster’s wireless connectivity and start navigating.

GPS Tracker WM-MFT-4G

Smart tracking. Total control. A rugged 4G GPS tracker for real-time fleet visibility, smart geo-fencing, and route intelligence — built for tractors, auto-rickshaws, construction equipment, trucks and two-wheelers.

Precision GPS tracking
<2.5 m positioning accuracy with continuous round-the-clock vehicle tracking.

4G LTE CAT-1 connectivity
Reliable and seamless real-time data transmission anywhere on the network.

Smart geo-fencing
Customisable virtual boundaries with instant entry and exit alerts.

10,000 location history logs
Route tracking, trip analysis and complete movement records on demand.

Ignition intelligence
Real-time ACC / engine ignition status monitoring and anti-theft security.

FOTA & SOS support
Remote over-the-air firmware upgrades plus SOS emergency button.

Tractors | **Auto-rickshaws** | **Construction** | **Trucks** | **Two-wheelers**



Technical specifications — WM-MFT-4G

GNSS	GPS + GLONASS · L1 · 33 tracking channels
Accuracy	<2.5 m CEP · tracking −167 dBm · acquisition −149 dBm
TTFF (open sky)	Hot start <5 s · cold start <40 s
Cellular	LTE CAT-1 4G module · Nano-SIM
Power	9–90 V DC input · 60 mAh / 3.7 V battery · battery backup
Interface	1 ignition input · 1 output (CAN optional) · LED: GNSS blue, cellular green
Physical	82.5 × 38.5 × 15.0 mm · 50 g · built-in GSM & GNSS antennas
Environment	−20 °C to +70 °C · 5–95% RH, non-condensing
Data storage	Up to 10,000 history logs (as per customer requirement)

ADDITIONAL FEATURES

LBS positioning
Cell-tower (station mode) positioning

Geo-fence
Optional customisable boundaries

FOTA updates
Firmware-over-the-air upgrades

CAN optional
For diagnostics or data exchange

Protocol
TCP (server based) and MQTT, customisable

SOS button
Enhanced safety in critical situations

GPS gateway

Reliable GPS gateway for industrial and logistics applications — multi-sensor monitoring, real-time tracking, telemetry reporting, and fleet management with secure cloud connectivity.

200 mAh battery backup

Internal backup battery ensures continuous telemetry transmission even if main power is disconnected.

Bluetooth & GPS connectivity

Dual wireless interfaces: precise GNSS location combined with BLE for wireless sensor integration.

Multi-sensor capability

Supports Security sensor, Fuel sensor, Temperature sensor, and Humidity sensor simultaneously.

Sends multiple parameters

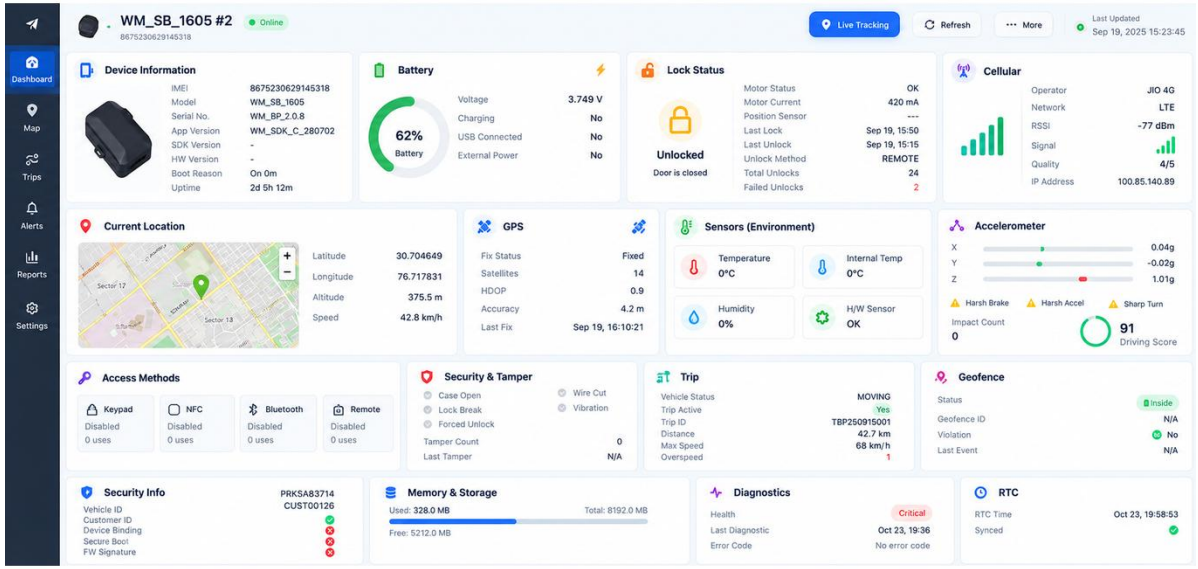
Transmits all telemetry parameters in real time for smarter, safer, and more efficient fleet operations.

Secure cloud gateway

Encrypted cloud communication to fleet dashboards and mobile monitoring applications.



Smart GPS & IoT Monitoring Dashboard



KEY FEATURES

1. Device Monitoring

Device info, firmware & health status.

2. GPS Tracking

Real-time location, speed & trip monitoring.

3. Connectivity

4G/LTE, Bluetooth and NFC interfaces.

4. Battery Management

Battery level, charging and power status.

5. Smart Lock Control

Lock/unlock status and access management.

6. Security Monitoring

Tamper detection, alerts & device security.

7. Driving Analytics

Harsh braking, acceleration & driving score.

8. Geofencing

Location boundaries & violation alerts.

9. Sensor Monitoring

Temperature, humidity & motion detection.

10. Remote Diagnostics

Device health, memory & error monitoring.

HW-EH13 — 4G Dual Camera Dashcam

Intelligent vehicle security dashcam with dual Full HD cameras, 4G LTE Cat 1 connectivity, live video streaming, and collision telematics.

Single Dashcam | **Dual Dashcam** | **Full HD 1080p** | **4G LTE Cat 1**

Dual Camera	Front & Rear/Cabin Recording
Video Quality	Full HD 1080p, 25 FPS
Connectivity	4G LTE Cat 1 module
GPS Tracking	Real-time vehicle location
Safety Features	Collision & vibration alerts
Storage	Up to 128 GB MicroSD
Audio	Built-in microphone & speaker
Power Supply	9–36 V DC wide input
Video Compression	H.265 / H.264 high efficiency
Compact Design	105 × 86 × 32 mm enclosure
Remote Monitoring	Video & data transmission
Applications	Fleet tracking, vehicle security & driver monitoring





Trusted by leading brands



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Hospitals, clinics and medical device companies

Defence & aerospace

Mission-critical electronics for defence

Banking & payments

POS terminals, soundboxes and payment devices

Pharmaceutical

Manufacturing equipment and monitoring systems

Energy & utilities

Smart metering, grid monitoring, industrial controls

Industrial & automotive

Industrial electronics, sensors and automotive

Walnut Technologies

Let's build what's next

Engineering precision with user-centric design — we welcome the opportunity to collaborate with your team.

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